

6/15/2025

INDUCTOR WINDING

Electromechanical Engineering Technology

ATMN-351 Industrial Automation Project

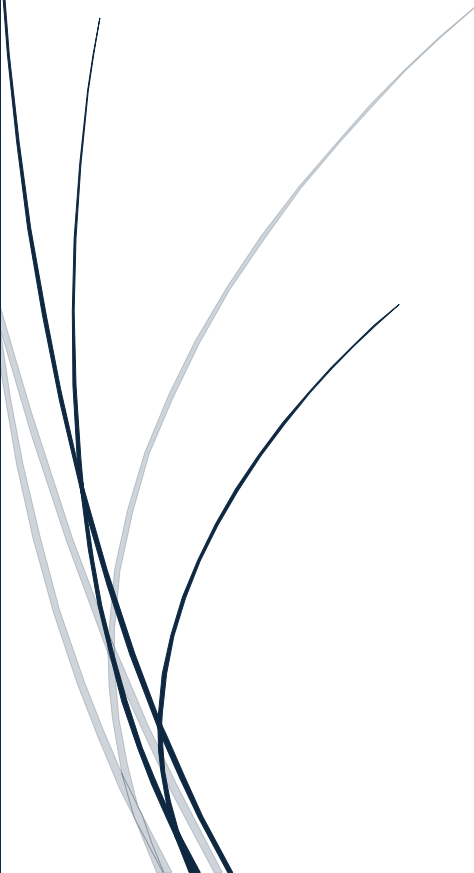
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Declaration of Group Authorship

I, **Paxton Coghlin**, confirm that this work submitted for assessment is my own and is expressed in my own words. Any uses made within it of the works of any other author, in any form (ideas, equations, figures, text, tables, programs) are acknowledged adequately at their point of use. A list of the references used is included.

I, **Grant Maddock**, confirm that this work submitted for assessment is my own and is expressed in my own words. Any uses made within it of the works of any other author, in any form (ideas, equations, figures, text, tables, programs) are acknowledged adequately at their point of use. A list of the references used is included.

I, **Rushawn Waite**, confirm that this work submitted for assessment is my own and is expressed in my own words. Any uses made within it of the works of any other author, in any form (ideas, equations, figures, text, tables, programs) are acknowledged adequately at their point of use. A list of the references used is included.

I, **Josh Le Blanc**, confirm that this work submitted for assessment is my own and is expressed in my own words. Any uses made within it of the works of any other author, in any form (ideas, equations, figures, text, tables, programs) are acknowledged adequately at their point of use. A list of the references used is included.

I, **Jaidan Mitchell**, confirm that this work submitted for assessment is my own and is expressed in my own words. Any uses made within it of the works of any other author, in any form (ideas, equations, figures, text, tables, programs) are acknowledged adequately at their point of use. A list of the references used is included.

Executive Summary

The following document is a report for the final year capstone project, which involved the automated winding of inductor coils. Group 1 comprises five members, each contributing to successfully designing and implementing the inductor winding system. Paxton Coghlin is the team leader, and PLC lead; Grant Maddock is the robot programming lead, PLC programming and mechanical design assistant; Rushawn Waite is the electrical lead, vision system assistant, and safety assistant; Josh Le Blanc is the Mechanical lead, vision system assistant and electrical assistant; Jaidan Mitchell is the safety lead and vision system assistant. This project incorporates a Fanuc robot which guides copper wire, assists in applying adhesive for coils, and places completed inductors at the pickup location. A programmable logic controller (PLC) manages the system's motion, including a winding mechanism, motors for feeding, and a pneumatic cutter for automatic wire termination. The load cells also provide additional information that measures the real-time spool mass. This lets us know how much wire is left to be wound. Additionally, we incorporated a Cognex vision system, which performs continuous quality inspections during operation. Safety is a key design component and is ensured by integrating dual SICK area scanners, interlocked spool cabinet doors, emergency stop buttons, and a reset system. The result is a fully functional cell engineered for precision, repeatability, and industrial safety, producing high-quality inductors with minimal operator intervention.

Hypothesis

This project will be a fully integrated inductor winding system. This system will utilize multiple subsystems, which will ultimately work in conjunction to manufacture a fully working inductor coil that can be applied to real electrical circuits and applications.

These components include a Fanuc robot, a camera, a brake line, a winding mechanism, and more. The project aims to provide more consistent, precise, and efficient manufacturing. The components will work together to minimize operator involvement and reliance. By monitoring the process in real time and enforcing the necessary safety controls through a PLC and vision system, the system is expected to reduce errors, downtime, and variability in production.

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Introduction

During the final semester of Humber College's Electromechanical Engineering Technology program, students were tasked with completing a capstone project as part of the Industrial Automation Project course. This project allowed students to apply theoretical and practical knowledge gained throughout the program and the hands-on experience acquired during work-related placements and other course-related projects. This project is expected to meet industry standards regarding both performance and reliability while operating within a controlled and safe educational environment. Group 1 was assigned a unique challenge, requiring integrating mechanical design, electrical systems, programming, robotics, and safety protocols. This capstone project focused on designing and automating an inductor winding machine to demonstrate how more practical and efficient automation can improve traditional coil manufacturing methods. Inductors are vital in various industries, including power electronics, automotive systems, and renewable energy. They are critical in filtering electrical noise, storing and transferring energy, and managing current flow in sensitive circuits. Despite the heavy use and presence of inductors, manufacturing them is often done manually. It is labor-intensive and heavily reliant on operator skill, increasing the chances of human error and making it more prone to inconsistency. The challenge was to create a repeatable, safe, and reliable system capable of producing inductors based on user input specifications, all while accommodating changes in inductance, wire gauge, and coil diameter. The system's core design centers around a Fanuc CR-4iA collaborative robot, which handles multiple tasks, including guiding the wire, applying adhesive, and placing the completed coil in the pickup location. A PLC coordinates all movement and logic,

including controlling a motor-driven winding mechanism, the wire feed roller, and a pneumatic-powered cutter for wire termination. System interaction is facilitated through a PanelView HMI, which allows the operator to control and monitor coil specifications and system status. The system also utilizes a load cell to monitor the wire spool and weight in real time. This tells us how much wire is left on the spool, helps ensure timely spool changes, prevents mid-cycle disruption, and helps maintain consistent production flow. Furthermore, a Cognex vision system is used to perform quality assurance inspections and to detect coil defects or improper winding, providing immediate feedback and enabling the system to halt operation for review before continuing the cycle. One of the significant challenges for this project was achieving the correct logic for the system's motion components. The robot, servo motor, and feed motor must operate in a carefully timed sequence to prevent over-tensioning or slack in the wire. First, the feed rollers advance the wire through a steel break line. The break line guides the wire to the cutting mechanism. This mechanism is a pneumatic cylinder equipped with a sharp metal blade mounted to its piston rod. After the wire is cut, adhesive is applied to hold its shape and ensure it stays securely wound during and after the coil formation. The robot then guides the cut wire into position, enabling the servo motor to rotate the winding mechanism to form the coil. This required careful calibration and tuning within the PLC logic and precise mechanical alignment of the winding and feed mechanisms. Managing this step without disrupting the overall flow demanded a reliable timing strategy and accurate robot positioning. In designing the cell, considerable attention was given to operator safety, given the presence of rotating machinery, pneumatics, and electrical systems. The workstation includes 2 SICK area scanners to

protect the open sides of the bench, a three-color stack light for status indication, and accessible E-stop and reset buttons. An interlocked cabinet door prevents access to the wire spool compartment while the machine operates. This project was designed to require as little human interaction as possible beyond initial setups, such as coil parameter specifications and spool replacement. This project bridges the gap between conceptual understanding and hands-on application, leveraging type hardware, control logic, and mechanical design principles. The outcome is a working prototype that simulates the real-world demands of precision manufacturing, preparing students for careers in automation, robotics, and advanced manufacturing environments.

Problem Statement

Inductors are used in many industries in various sizes and capacities. From filtering/smoothing high-frequency noise in electrical circuits to storing and transferring energy in power converters, inductors play a role in many major industries. However, making inductors with a reliable and repeatable process was a problem for our customers. The customer wanted inductors of varying inductances, wire gauges, and diameters. All these parameters should be able to be input into the system, and the system would produce the inductor coil autonomously based on the specifications. Some of the bigger challenges will be to match the speeds and synchronization of all the systems involved, as the system is meant to be integrated. Using robots, servo motors, and DC feed motors requires them to work together to achieve a well-manufactured product. The system requires a streamlined approach that produces a coil in a smooth, autonomous process with as little human interaction as possible, from start to finish. This ensures that production can be maximized while keeping the cost of each unit to a minimum.

Hypothesis

This project will be a fully integrated inductor winding system. This system will utilize multiple subsystems, which will ultimately work in conjunction to manufacture a fully working inductor coil that can be applied to real electrical circuits and applications.

These components include a Fanuc robot, a camera, a brake line, a winding mechanism, and more. The project aims to provide more consistent, precise, and efficient manufacturing. The components will work together to minimize operator involvement and reliance. By monitoring the process in real time and enforcing the necessary safety controls through a PLC and vision system, the system is expected to reduce errors, downtime, and variability in production.

Roles and Responsibilities

Paxton Coghlin:

My primary role in the capstone project was as team leader, a position that required adaptability, initiative, and a strong focus on communication and team cohesion. I approached the leadership role with the mindset that a team lead's responsibility is to fill in gaps where others may be struggling, help where needed, and ensure that everyone remains motivated and aligned with the project's end goals. To maintain consistent progress, I scheduled team meetings twice a week. These meetings were coordination sessions and dedicated work periods, enabling team members to discuss challenges, share updates, and receive prompt feedback.

As the semester progressed and most of the team was working in person, we found it effective to reduce the frequency and duration of scheduled meetings, as we would often be together working on the project anyway. Our documentation and file management were handled through Microsoft Teams. Due to significant overlap in responsibilities, particularly in mechanical design, assigning one person to manage all files would have been inefficient. Instead, we organized our files by discipline (e.g., mechanical, electrical, programming), which allowed team members to work collaboratively and step in seamlessly when others needed support. I emphasized good file-naming and folder structure practices to ensure any team member could navigate the project files without confusion.

I also took a strategic approach to assigning roles during the project planning phase. I encouraged each team member to take on tasks outside their previous co-op or

academic experience. My goal was to ensure that the capstone remained a valuable learning opportunity for all involved, rather than simply repeating familiar work. This also fostered a collaborative environment, where someone with experience in each area could support others' learning.

In addition to leadership, I served as the lead for PLC programming. While I had extensive mechanical experience, several other group members had strong PLC backgrounds, so this role allowed me to expand my skills. I was responsible for programming the PLC and managing communications between the controller and peripheral systems, including the area scanners, Cognex vision system, HMI, and industrial PC. We used Ethernet/IP for its simplicity and our team's familiarity with the protocol.

Initially, we planned to use an OPC server to move data directly from an Excel spreadsheet to the PLC, but licensing limitations with RSLinx forced us to find an alternative. I wrote custom Python code that runs on the industrial PC, enabling seamless data transfer from a .csv file into the PLC's memory via Ethernet/IP.

I was also heavily involved in the mechanical systems. I designed the winding mechanism and carried out much of its ongoing maintenance. Initially, the mechanism used two servo motors, but I redesigned the system during the build phase due to budget constraints. The revised design uses a single servo motor and a solenoid, which locks the rotating core to the top or bottom, allowing for diameter adjustment and part removal. Additionally, I converted the actuation system from pneumatic to electric to eliminate the need for an expensive pneumatic slip ring. The electric actuators provided

a cost-effective and reliable alternative. I also contributed to the design of the extruder system. I developed the preliminary concept, which my teammate Josh later refined and improved upon, resulting in a functional and efficient final design.

On the electrical side, I contributed to both troubleshooting and design. I was solely responsible for the electrical design of the adhesive extruder. After our initial approach was ruled out due to cost, I designed a low-cost alternative using two heating elements (\$20) and two thermistors (\$1.60), with a potentiometer sourced from a drawer somewhere. This solution brought the total component cost to just over \$21 and successfully met our functional requirements.

Throughout the project, I played a role in nearly every discipline, including mechanical, electrical, and controls, sometimes as a lead and other times in a support capacity. My contributions reflect a strong commitment to the project's success and the development of my skills and my teammates.

Grant Maddock:

My role was to program the Fanuc robot and the HMI and integrate both systems into the PLC code. I was also responsible for designing the wire feed system. This was my first experience with Fanuc programming, so there was some difficulty initially, but Rushawn was very helpful in steering me in the right direction. The robot was programmed to pick up the wire from the cutting station, insert it into the winding mechanism, guide the wire as the coil is wound, and then pick up the completed coil from the mechanism, placing it down for an operator to grab it. The robot's main challenge was working with the unusual end-of-arm tooling. The end-of-arm tooling

consisted of a Robotiq gripper, a hot glue applicator, and mechanical guides. At first, there wasn't much support for the wire, so I designed a docking station for the robot at the outlet of the cutting mechanism. This new docking station positioned the wire more consistently, allowing a smoother transition into the winding mechanism.

When designing the HMI, I built it in three main sections. The first section is the "About" section. This section is available to anyone who accesses the cell and explains what our cell makes, the team that built it, and our responsibilities in its completion. The next main section is the "Control" section. The operator uses these screens during normal working conditions. The operator has access to the start, stop, and reset buttons, feedback on the process, and a reload screen for when the wire needs to be restocked. The last section is reserved for engineers and technicians. The "Manual" section provides complete control of the cell for troubleshooting and testing purposes. I designed the HMI with a charcoal and copper colour scheme to complement the copper wire we use in the system. I used consistent radii and a simple sans serif font, inspired by modern "dark mode" web design.

Communication between devices was done using Ethernet from the HMI to and from the PLC. Since Allen Bradley made both devices, the communication was easy to set up; I just needed to ensure that tag names were consistent across both devices. The robot was a bit more challenging to set up for communication. This was also done over Ethernet, and I communicated with the PLC by passing several binary bits. These bits notified the PLC when the robot was in a specific position, and the robot would wait until the PLC sent a bit telling it to start the following action.

When designing the feed mechanism, I first set three criteria: the mechanism needs to support multiple wire gauges, work with minimal input from the operator, and consistently guide the wire to its destination. The solution I came up with consisted of steel tubing and counter-rotating polypropylene wheels. The wire is first fed off spools in the cabinet and into the wheels. The wheels grip the wire and push it through the steel tubing, which guides it to the cutting mechanism. The polypropylene wheels conform to the shape of the wire, allowing it to feed a wide variety of gauges, and the steel tubing prevents any unwanted movement from the wire. Inductive proximity sensors were installed along the tubing to track the wire's progress as it feeds through.

Rushawn Waite:

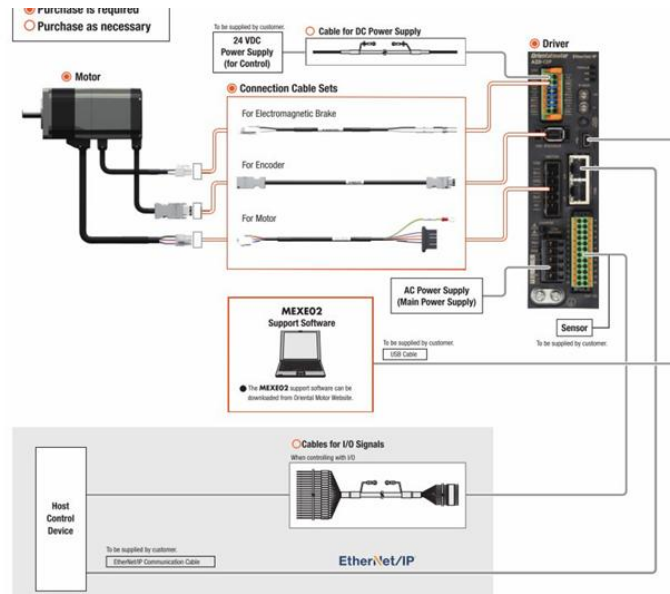
As the Electrical Lead, my primary responsibility was designing, wiring, and integrating all electrical components of the machine. This included power distribution, sensor input wiring, actuator control, safety system wiring, and device communication over an Ethernet/IP network. I ensured all devices were connected correctly, powered, and interfaced with the PLC or associated controllers. Josh also helped with some of the troubleshooting and testing of components.

Load Cell Wiring: The load cells used for tension feedback were connected to the analog input card of the Compact Guard-Logix PLC. I routed them separately from high-power wires to reduce electrical noise. I also ensured proper grounding and terminal labeling to assist with troubleshooting.

Temperature Sensor: A temperature sensor monitored the glue extruder's heating block. I wired it to another analog input channel to get constant feedback on the hot glue extruder's temperature. The heating elements were controlled through solid-state relays, which I wired to the Armor-Block outputs module. I used proper heat-rated wire and added inline fuses for protection.

DC Motor Control: Three bidirectional DC motors were each controlled using two relays configured as an “H-bridge.” I wired the relay coils to the PLC output card. Using the relay's contacts, I wired the control so that if one of the relays is energized, the motor turns forward or backward, and if both are energized, the motor can't be powered. These motors are part of the feeding mechanism. (Figure 11. DC Motor Wiring).

Servo Motor and AZD-AEP Driver: I wired the AZD-AEP driver to 120V AC power for the motor and 24VDC to power the drive. The communication lines were connected to our Ethernet switch to enable PLC communication. I also connected the motor's encoder and control lines to the driver, following the manufacturer's wiring diagram and safety guidelines. Figure 1. Servo Motor and AZD-AEP Driver.



Start and Reset Push Buttons: The momentary Start push button is wired to the Input Armor-Block module, and the Reset button is wired to the PLC's safety input card. The reset button is part of the safety system; hence, it is placed on the safety input.

Safety Devices: I installed and wired the following safety components:

- E-stop button (two-channel): wired to the PLC safety input card channel 0.
- Door interlock switch: hardwired to the safety input channel 1 to stop all motion when open.
- Area scanners (microscan3 and nanoscan3): The nanoScan3 OSSD 1 is wired to channel2 of the safety input. The microScan3 bits are passed over Ethernet.

I used an Omron safety relay to integrate the Fanuc robot into the safety circuit.

The robot's external E-stop circuit was wired to the relay's normally open

contacts. The relay's coils (K1 and K2) are controlled by outputs from the PLC's

safety output card. This allowed the robot to halt safely during any E-stop event.
(Figure 12. Safety I/O).

HMI and Industrial PC: I connected the HMI to the PLC using an Ethernet cable routed through a managed switch. The HMI was powered by the 24V DC control supply and appropriately grounded. The Industrial PC was also powered from a protected supply and connected to the same network, allowing it to receive and send data.

Robotiq Gripper Controller: I wired the Robotiq controller to the gripper on the robot's flange. The 24VDC power supply powers the controller. It is connected via Ethernet to the switch, and I confirmed communication with the Robot to trigger open/close functions during pick-and-place operations.

Cognex Vision System: The Cognex vision camera is mounted over the work area and powered using 24VDC wiring. I connected the Cognex to the Ethernet switch and tested communication to ensure the PLC could use the vision system feedback.

Solenoid Valve (Cutting Station): I wired one solenoid valve from the 8-station manifold to a digital output on the Armor-Block module. This solenoid controls a cylinder that extends and retracts the blade of the cutting station.

Challenges and Problem Solving

As part of my role as Electrical Lead, I encountered several technical challenges during the wiring and integration of components. These challenges provided an opportunity to apply troubleshooting techniques and electrical principles in real-world conditions.

Load Cell Signal Not Detected by PLC: After wiring the load cell sensors to the PLC's analog input card, I observed no data had been received. I used a multimeter to check the output voltage from the load cells and discovered that the voltage was extremely low, below the minimum threshold required for the PLC to register an analog input.

Solution:

I added signal amplifiers between the load cells and the PLC to solve this issue. These amplifiers boosted the signal voltage into the acceptable range for the PLC's analog input, restoring reliable tension data from the sensors.

Solenoid Overheating: One solenoid used in the cutting station was rated for 300 mA at 24VDC. However, when wired to the system and activated, the solenoid became extremely hot quickly, raising safety and reliability concerns.

Initial Troubleshooting:

- I attempted to reduce the current by wiring resistors in parallel, but this caused the solenoid to fail to energize properly.
- I then experimented with 12VDC and 24VDC solenoids, installing a dedicated 12VDC power supply and trying different configurations.

- Despite the efforts, the solenoids either failed to energize reliably or continued to overheat.

Root Cause and Final Solution:

Upon further investigation, we discovered that the solenoid was a quick-fire design, intended to operate only for short bursts, not continuous duty. It was unsuitable for how we intended to use it (as a continuously energized actuator).

To solve the issue, we physically inverted the solenoid. This allowed it to function in reverse logic; it now engages when de-energized and disengages only briefly when energized. This change eliminated the need to energize it for extended periods, preventing overheating and preserving functionality.

DC Motors Running Too Fast: I installed two bidirectional 50 RPM DC motors to control the limiting and clamping mechanisms. However, during testing, the motors rotated too quickly, which could have caused mechanical damage or misalignment.

Solution:

I implemented an inline DC motor speed controller with each motor and used relays to manage direction control (forward/reverse). After tuning the speed controller output and coordinating it with relay switching logic in the PLC, the motors operated at the correct speed and in both directions as required.

Josh Le Blanc:

My role and responsibilities as mechanical lead were primarily to oversee the design and mechanical build of the inductor winding system. During the first design semester, I was tasked with creating a layout and simple designs of what the system should look like and how it should function in the long term. I started with layout sketches and gradually transferred my ideas to CAD drawings.

Their complete assembly consisted of many smaller, custom-made parts that required a lot of intricate design. Because of this feature, we opted to 3D print many of our parts where we could reduce the material cost and build time. The major mechanical components consisted of the following subassemblies:

Overhead Enclosure- Originally, this was supposed to be a fully enclosed custom extrusion cell with 40x40mm extrusion as the frame material, clear acrylic, and light curtains on the open side to allow for interaction with the cell. Due to budget constraints, it was reduced to a 28x28mm extrusion. Then, the whole assembly was redesigned from the ground up only to make the cabinet for the wire spools, which was enclosed with supports for acrylic holding the HMI and control buttons. Safety was converted from physical barriers to area scanners as the first defense.

Winding Mechanism- Paxton did the main design. I dealt with most of the minor modifications that made it functional once it was designed. It proved to be one of the more complex parts of the system to assemble, as it had multiple parts that moved together and independently at times. The most significant setback with the Winding was that the solenoid we had kept getting so hot that it would melt the 3D print, and once

even burned itself out. We eventually solved the heating issue by inverting the solenoid, reducing how long it would take.

Feeding Mechanism- Grant initially designed this system to feed the wires from the spools to the system. I went on to make minor edits in the design and machine the shafts needed for the motor to transfer power. This required the first creation of the actual drawing files for components so that I had dimensions and references. I also had to add tolerances and ensure the parts fit correctly. It was also a good refresher of the lathe, which I hadn't used in months.

Cutting Mechanism- Jaidan was tasked with designing a cutting mechanism that worked, as this was a big issue a previous group had faced with their custom cutter never working. After a few revisions and consultations, we used a big-bore cylinder for maximum force and a razor blade as the final edge. Not only did it work very well, but it also allowed us to change the blade out for easy maintenance.

Adhesive Extruder- Paxton did the initial design but underwent heavy modifications by me to conserve space and weight, as I later decided to attach it to the gripper as an extension of the robot's end effector. Due to the nature of this part getting very hot to melt the hot glue stick that it used to stick the coils together, most of the parts had to be machined out of aluminum and steel for better heat conduction, and in other cases, better heat dissipation away from the gripper itself. Much of my mid-semester time was spent machining parts for this assembly.

The other small assemblies and miscellaneous parts we fabricated along the way were added to the remaining subassemblies. Most of my time in the project's building phase

was spent assembling and machining parts, redesigning, and dimensioning drawings for new and existing parts.

On top of my mechanical responsibilities, I also spent a lot of time reading up on documentation for the various components in the system. I did a lot of background integration, like configuring and commissioning devices, setting up and assigning IP addresses, and network communication for the programming base of the project. Along with Jaidan, we set up a baseline safety program with the area scanners and the vision system, followed by the integrated program and communication with the PLC. Grant, Rushawn, and I did robot commissioning, setup, and PLC communications.

I assisted Rushawn with troubleshooting, testing, and wiring where I could, for electrical installations and I/O testing of the system before any PLC logic existed, to ensure everything was configured and communication was proper. Also, I ensured he helped get documentation and find wiring diagrams when clear instructions were not given. In the latter half of the project, I also helped with PLC logic getting started and base programs, getting components we had built working and integrated into the system.

Many of my challenges stemmed from having many custom parts that had to be machined. I spent many afternoons and nights lathing, milling, and cutting in between classes and after hours to ensure things were built to specification and were functional. Towards the end of the project, having a lot of custom and highly detailed drawings proved to be the most difficult of all my tasks, ensuring that my as-built drawings were up to date and editing drawings I had previously done.

Jaidan Mitchell:

This semester, my role in the inductor winding project evolved from mechanical development to a more technically demanding position involving safety and vision system integration. Last semester, I focused on initial mechanical component design, such as helping develop the original wire cutting mechanism and researching gripper designs. I integrated and configured the vision system, updated area scanner safety fields, and assisted in general troubleshooting and system refinement. One of the main areas I contributed to this semester was configuring the Cognex vision system. This device was used to verify coil quality during the manufacturing process. Josh helped me configure this device by helping adjust the detection parameters and troubleshooting inconsistencies in the vision tool's readings. One of the main areas I contributed to this semester was configuring the Cognex vision system. Josh helped me set up this device. This was used to verify coil quality during the manufacturing process. However, during configuration, we quickly encountered problems as the Cognex wasn't correctly detecting the wires due to poor lighting and contrast. The camera frequently failed to identify incorrect wire placement or flagged valid setups as errors. After testing several potential solutions, this issue was eventually resolved by enhancing the lighting in the camera's viewing area. This was done by mounting a flashlight on top of the Cognex. This light only turns on when the wire is ready to be inspected, ensuring consistent illumination without causing unnecessary glare or interfering with other parts of the process. By synchronizing the flashlight with the inspection stage, we significantly improved detection accuracy while keeping the vision system efficient and focused. This significantly improved the reliability of the vision system and allowed it to perform more

consistent inspections. From that point forward, the system could detect wire configurations consistently. This process gave me valuable hands-on experience in vision automation, a field in which I previously had little exposure to. Through troubleshooting issues, working with industrial software like In-Sight Explorer, and experimenting with physical setups such as lighting alignment and camera positioning, I developed a much stronger understanding of how vision systems are integrated into automated processes. Alongside the Cognex system, I also helped configure the SICK area scanners, which were critical in maintaining operator safety. Rushawn aided me throughout this process; he handled the electrical wiring of both scanners since he was the electrical lead. Once powered and connected, we worked to redefine the scanner's safety fields using SICK Safety Designer software, especially after changing the cell's layout. This included adjusting the warning and protective zones, repositioning the scanners for optimal coverage, and verifying that the system would reliably trigger safety shutdowns if the defined zones were breached. Our original scanner layout became outdated once we were forced to reposition the robot and winding mechanism due to resistance in the brake line. With the robot, cutter, and winding assembly moved to a new location, the safety fields defined in SICK Safety Designer had to be completely redone. This reconfiguration included adjusting the angle of the scanner and the mounting position to cover the new layout adequately. I also edit field shapes, verify scanner overlap, and test live responses with the PLC. I also worked closely with group members to ensure both scanners were coordinated and that their outputs correctly triggered system shutdowns when safety zones were breached. In addition to my technical contributions in safety and vision system integration, I undertook several

hands-on fabrication tasks throughout the semester. Furthermore, I assisted Josh in fabricating the metal block intended to mount on the cylinder's piston rod for the cutting mechanism. This involved operating a bandsaw to help accurately cut the block to the required dimensions. Precise cutting and shaping were critical to ensure clean wire termination and smooth pneumatic actuation. In addition, I helped operate a lathe to machine deeper holes into the block, allowing it to securely interface with both the piston rod and the fixed cutting surface. This ensured proper alignment and stability during cutting operations, essential for consistent and reliable wire shearing. These fabrication tasks gave me valuable experience with basic and intermediate metalworking tools. They highlighted the importance of physical accuracy when translating digital designs such as SolidWorks into real-world assemblies. Beyond my responsibilities, I also provided general support across various project parts. During the final integration phase, I assisted with wiring, mechanical alignment, and testing tasks. I frequently coordinated with other team members to help resolve layout changes, sensor field mapping, or mechanical interference issues. This semester marked a significant step forward in my technical and teamwork abilities. I gained hands-on experience configuring a vision system, reconfiguring safety scanners, and solving integration challenges under tight deadlines. These tasks highlighted how engineering requires constant adaptation across mechanical, electrical, and software systems. Working closely with team members taught me the value of clear communication, especially when coordinating changes that affected multiple subsystems. My main contributions in vision integration and safety setup supported the system's reliability and helped me grow into a more well-rounded and collaborative engineer.

Methodology

The methodology outlines the structured engineering approach to designing, implementing, and testing the automated inductor winding machine. The development process followed a modular workflow, beginning with CAD and schematic design, mechanical modeling in SolidWorks, electrical panel construction, and control system programming. Each subsystem was built using industrial-grade components and was engineered with a focus on performance, safety, and reliability. Testing and implementation occurred during the build phase, where individual subsystems such as the wire feed, cutter, winding assembly, and vision inspection were first validated independently. These components were then integrated into the complete system in stages. This method allowed the team to identify and resolve functionality issues early, ensuring smooth interaction between all components and achieving a reliable, consistent, and safe operational workflow.

Load Cell Testing

We began verification by testing the load cells integrated into the spool holder. These sensors measured the weight of the wire spool to determine how much wire remained. We tested this mechanism by placing different spools on the holders and confirming that the load cells produced accurate and consistent readings across various coil sizes and wire gauges.

Feed Mechanism Testing

The feed roller mechanism, which pushes and pulls the wire through the system, was tested for accuracy and responsiveness. We verified that the rollers maintained proper tension and smoothly fed the wire through each stage without slipping or jamming.

Wire Cutter Testing

The wire cutting mechanism was tested to ensure it consistently delivered clean, accurate cuts at the correct point in the cycle. We confirmed that the pneumatic cutter triggered reliably and could handle multiple wire gauges without damaging the coil or leaving excessive wire behind.

Glue Extruder Testing

We tested the glue extruder to confirm it dispensed the appropriate amount of adhesive after the wire was cut. The tests verified consistency in glue application, strong adhesion to the wire surface, and safe operation throughout the heating and cooling phases of the process.

Winding Mechanism Testing

The winding mechanism was tested using various parameters to validate its ability to accommodate different coil diameters and lengths. We confirmed that the wind was uniform, secure, and properly aligned with minimal variation between cycles.

Fanuc Robot Testing

We conducted repeated tests to confirm that the Fanuc CR-4iA robot operated with high repeatability. We verified that the robot properly guided the wire to the winding position,

interacted correctly with other mechanisms, and maintained precise motion without collisions. It functioned safely and coordinated with other systems as programmed.

PLC Testing

The PLC control system was evaluated for correct sequencing, real-time decision-making, and responsiveness to sensor input. We confirmed that the logic reliably coordinated all components and executed each process step as intended.

Vision System Testing

The Cognex vision system was tested for its ability to detect winding errors and irregularities in real time. The system successfully flagged defects and responded appropriately by halting the process or notifying the operator via the HMI. Its performance was confirmed both independently and within the integrated system. Each component underwent individual and integrated testing to ensure complete system readiness and autonomous operation.

Electrical

The electrical subsystem was developed through a structured process combining circuit design, hardware research, and hands-on implementation. During the initial phase, detailed schematics were created using AutoCAD Electrical, incorporating all control and power components for the inductor winding system. Extensive research was conducted to determine the proper configuration of each device, including sourcing vs. sinking logic, voltage and current specifications, safety ratings, and signal types, to ensure compatibility with the safety-rated PLC, servo drives, and sensors. As of this stage, most electrical components have been installed, wired, and labeled adequately

within the control panel, with only a few remaining connections to be completed.

Devices are mounted and connected using 24 VDC and 120VAC wiring, and wires have been color-coded and routed through cable tracks for organization and maintainability.

Relay modules create bidirectional motion using the PLC with DC motors that drive the copper wire feeding mechanism, allowing safe forward and reverse motion control.

Each device is individually tested upon installation throughout the wiring process to verify functionality and signal integrity before proceeding. This includes checking polarity, voltage presence, I/O state transitions, and safety device behavior. Adjustments to the design were made where necessary due to hardware substitutions or missing components, and all changes are being updated in the AutoCAD Electrical schematics.

Mechanical

The mechanical component of this project began with basic ideas, sketches, and flowchart diagrams. From there, we were able to break down these basics into more detailed drawings and even subsystems that needed to be done based on the system's needs and the customers' wants. Getting a preliminary Bill of Materials helped us decide what components to use structurally within our budget and what would be cheaper and easier to buy versus fabricating or machining ourselves. Once these were sorted, mechanical drawings were made to understand better and model these aspects and to make them easier to build. Effectively, we broke down the building into a few main subsections mechanically.

1. The main frame consists of the extrusion frame mounted on the cell base.

Initially, it was meant to be a fully enclosed frame made from 40x40mm extrusion.

However, due to costs and import limitations, it was redesigned to use in-house

28x28mm extrusion. Then, realizing that this too would be ineffective due to acrylic cost and size restraints, we opted to use sections from a previous project with modifications that would save time and money on the building but still allow functionality and preserve the integrity of the project.

2. Winding Mechanism: This component was custom-made and intricately designed to suit the project's needs. It encompassed the ability to change the diameter of the coil by incorporating a lathe chuck design that opens and closes, changing the diameter. Initially, it was meant to use two motors, but it had to be redesigned to use one due to budget restraints. This was mitigated using a solenoid to engage and disengage the varying lathe chuck and the rest of the winding mechanism that allows the entire base to spin, winding the actual coil.
3. Feeding Mechanisms: The feeding mechanisms dispense the various wire gauges to the winding mechanism. They consist of a motor driving a soft rubber wheel, which drives another similar wheel supported by bearings. The wire is then drawn through between the wheels. This component required a few revisions to meet the needs of the different gauges, including one design change and a motor change. (Figure 9. Feeding Mechanism).
4. Cutting Mechanism: This terminates the coil when winding is done. It consists of a pneumatic cylinder attached to a block with a razor blade that chops the wire against it. Our biggest issue was getting the wire to be terminated cleanly in one go without deforming the blade or not cutting the wire all the way through. (Figure 9. Feeding Mechanism).

Overall, the mechanical required the aid of various group members, ensuring that the components worked and did their job.

Programming

The PLC programming component of this project played a crucial role in coordinating and automating all subsystems to function as a cohesive unit. Development began by outlining the control logic using flowcharts and pseudocode, which helped identify the required key inputs, outputs, and sequence of operations. From there, the team developed the PLC program using ladder logic, SFC, and structured text, using each “language” for its respective advantage.

Programming efforts focused on building reusable routines for each subsystem and organizing the logic into clearly defined states. This made the system easier to debug and maintain while allowing for adjustments as the mechanical and electrical designs evolved. Writing the program in a highly parameterized fashion helped react to the changes that naturally come with project progression.

Key functions within the PLC program included:

Winding Control Logic: The winding mechanism’s logic was written to monitor its current position and automatically control motor speed and direction to get to a new requested position. Alongside a solenoid at specific intervals for chuck engagement, the winding mechanism's positions can be easily changed. Timers and counters were used to ensure consistent winding cycles, and interlocks were added to prevent engagement during unsafe conditions.

User Interface and Monitoring: An HMI was integrated with the PLC to allow users to start, stop, and reset the system and view real-time system status. The interface also allowed users to adjust parameters such as coil size, wire gauge, winding speed, and cut timing, improving usability and flexibility.

Order Management: An additional software component running on an industrial PC handles the automated transfer of inductor “orders” into the PLC’s memory. This software reads order data from a spreadsheet and monitors the PLC for a signal indicating readiness to begin a new job. When triggered, it sends the relevant parameters for manufacturing the next inductor. This setup allows the PLC to retain control over the process flow while enabling seamless, automated order handling from an external source.

Throughout the development process, extensive physical testing was carried out to fine-tune the timing and reliability of each sequence. Debugging and refinement were done collaboratively, ensuring each function worked smoothly in the whole system context. The result was a reliable, fully automated control program that enhanced the mechanical building's precision, repeatability, and safety.

Safety Systems

Area Scanner

To ensure complete perimeter protection of the workstation, the system incorporates a SICK microScan3 Figure 2. Microscan3 Safety Fields. And a nanoScan3 area scanner, Figure 3. (Nanoscan3 Safety Fields). These devices are mounted to protect the open sides of the workbench and provide overlapping coverage for redundancies. Each

scanner monitors a configurable safety field and continuously scans for any intrusions. If an obstruction is detected, both scanners' output switches turn off, halting all system movement. The combined use of both scanners ensures robust and fail-safe protection of the entire cell during operation.

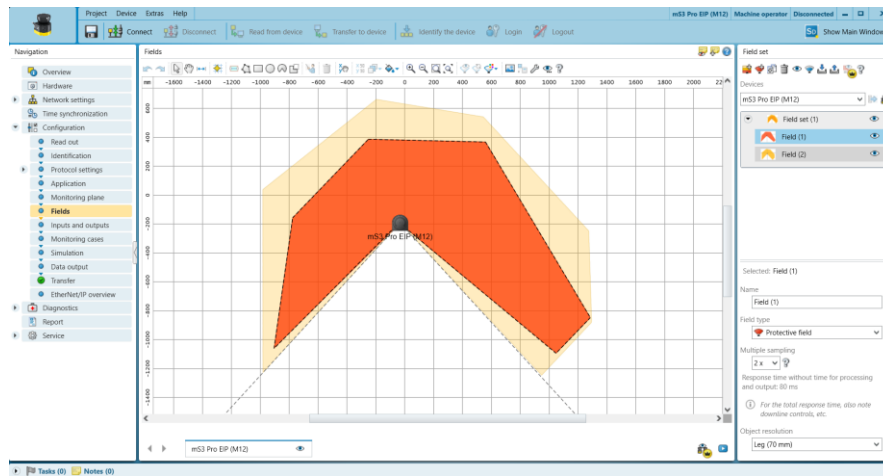


Figure 2. (Microscan3 Safety Fields)

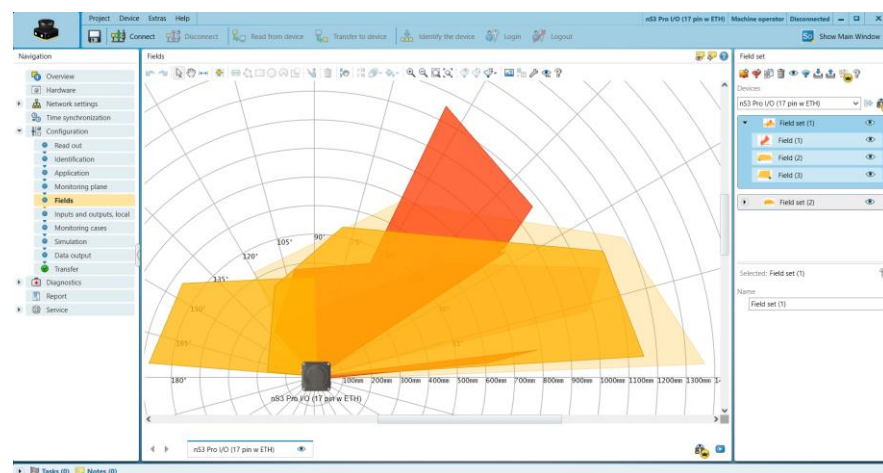


Figure 3. (Nanoscan3 Safety Fields)

Stack Light

In addition to the area scanners, the stack light provides visual feedback on the system's safety status. These three colored lights serve as a real-time indicator for operators, helping them quickly assess whether the machine is safe to operate. When the light is green, it signals that the system is in a normal operating state and is ready to run. A yellow light indicates that the area scanner has detected an object within the warning field, while the system may still be running; caution is advised. If the light turns red, an object has breached the protective field, prompting the scanner to deactivate the OSSD outputs and initiate a complete safety halt of the system's components. In this state, the system cannot be operated until the obstruction is cleared and the reset button is pressed to re-enable functionality. This visual alert system reinforces safety by communicating any potential hazards clearly and promptly to users.

Reset and Emergency Stop button.

This system is equipped with an emergency stop and reset button. This allows the operator to immediately shut down all system activity in case of a hazard or malfunction. When pressed, it overrides all other inputs and locks the system until the issue is resolved. After the E-stop is released, the reset button must be pressed, and safety conditions must be verified to resume operation. This manual override mechanism ensures operators can fully control the system's safety, even in unexpected scenarios.

Project Difficulties

Our team encountered various technical and logistical challenges throughout this project, forcing us to adjust our original design significantly. These obstacles ultimately led to a more refined and practical solution, but also required extra time, troubleshooting, and coordination.

Winding Mechanism

One of the earliest difficulties we faced was with the winding mechanism design. The initial concept included two motors. One would rotate the coil, and another would operate like a chuck-style opening mechanism. However, we abandoned this approach due to budget limitations and re-engineered the system to function with just a single motor. To solve this, we incorporated a 24VDC solenoid, which acted as a clutch to engage and disengage the winding chuck. This solution allowed us to achieve the necessary mechanical function without needing a second motor. Still, it also meant redesigning the internal assembly of the winding platform and adjusting our control logic to accommodate the solenoid's timing and actuation requirements. (Figure 6. Winding Mechanism).

Vision System

Another problem we had was with our vision system. During testing, the Cognex vision system experienced issues with inconsistent detection. One of the main challenges was that the spool wire often blended visually with the surrounding winding mechanism due to poor lighting and reflective surfaces. These conditions created glare and reduced contrast, making it difficult for the vision tool to distinguish between different wire configurations reliably. As a result, the system frequently produced false negatives and positives, such as failing to detect improper wire setups and identifying correct configurations when none were present. Our initial attempt to improve detection involved pulling down the curtains in the capstone room and maximizing ambient lighting, but this had little effect. We then tried adjusting the camera's internal settings in In-Sight Explorer, including brightness, exposure, and contrast. While we noticed slight improvements, the results remained unreliable. Eventually, we addressed the issue's root by physically enhancing the lighting conditions. We mounted a flashlight onto the same extrusion as the Cognex camera, aligning it to illuminate the camera's field of view directly. This simple addition drastically improved visibility and consistency, allowing the vision system to identify wire configurations with much greater accuracy.

Wire Feed Path and Layout Redesign

As we began final system integration, we encountered a significant issue with the wire feeding process. The wire was stalled by excessive resistance in the brake line due to its long and curved path to travel from the spool cabinet, around the robot, past the cutting mechanism, and finally into the winding system. The combination of distance and tension made it difficult for the wire to maintain enough momentum, and in most

cases, it wouldn't even reach the gripper for clamping. This breakdown in flow forced us to re-evaluate the entire configuration and made it clear that our layout was no longer viable despite our limited time. We had to completely reposition the robot and relocate the cutting and winding stations closer to the spool cabinet. By doing this, we reduced the wire's travel distance, eliminated unnecessary bends, and minimized tension caused by routing. However, this wasn't a simple shift; it required us to reteach all the robot's programmed points, recalibrate its motion paths, and test to ensure it could still reach and guide the wire into the winding mechanism correctly without collisions or clearance issues.

Safety Scanner Reconfiguration

Due to the revised robot layout, this last-minute overhaul required us to reconfigure our safety fields. We used two SICK safety scanners, one of which was a microScan3 and the other was a nanoScan3. These devices covered the perimeter of the cell. However, since the physical workspace changed, the original safety zones no longer matched the robot's reach and motion paths. This forced us to redefine the scanner fields in SICK Safety Designer software and adjust the scanner angles and positions to ensure that the warning and protective zones effectively covered the new layout. Ensuring that the scanners overlap and maintain full coverage while avoiding blind spots was a delicate process that required multiple rounds of live testing.

Wire Spool Limitation

In addition to the mechanical and layout-related issues, we faced another setback with wire spools. The original plan was to integrate a system capable of handling three

spools to demonstrate a broader range of inductors and wire gauges. However, we could only operate one spool by final testing due to time constraints, hardware limitations, and the complexity of managing multiple feed paths within the cell. This reduced the system's overall flexibility and meant we couldn't fully showcase the multi-spool automation we had initially aimed for. A compromise had to be made late in the project to ensure that at least one complete cycle could be reliably demonstrated.

Wire Field Attenuation Strength

Inductive sensors detect objects by sensing changes in their electromagnetic field. In our case, the wire used to wind the inductors didn't attenuate the field enough to trigger an "on" signal. After experimentation, we found a workaround: taping small pieces of aluminum foil to the sensor's sensing face. This brought the sensor closer to its activation threshold, allowing the wire's presence to push it over the edge. Fine-tuning was tricky; too much foil would keep the sensor always on, and too little wouldn't help. Several attempts were made to get the balance right due to the many small variables involved.



Figure 4. Aluminum foil on tape, ready to be wrapped around the sensor

Solenoid Issues

When swapping from multiple servos to a solenoid as a clutch for our winding mechanism, we used one of the extremely common blue 24V solenoids. Because of their prevalence, it is hard to trust any single datasheet because manufacturers often lie on them to get the sale. The CAD files we found on the internet were incorrect, causing parts in our mechanical building to collide. These solenoids are designed to be on momentarily, not constantly on. Combining these factors caused us to destroy five solenoids before finally having the sixth solenoid work.

Final Integration Adjustments

These setbacks and design changes meant we had to constantly update our SolidWorks assembly, re-test electrical wiring, and revisit PLC code throughout the build. While this approach slowed our initial progress, it led to a more reliable and practical machine. Despite the challenges, these modifications allowed us to meet the project goals better and deliver a working solution closer to what would be expected in a real industrial application.

Proposal Modifications

One of the original ideas for this project was to incorporate a spreadsheet interface that would allow the operator to input key parameters such as wire gauge, coil diameter, and target inductance values. The PLC would have read this data to automatically configure the system for a specific coil design, helping streamline set-up and reduce the chances of human error. The vision was to combine this input with built-in error-checking functions to provide alerts if the entered values fell outside acceptable operating ranges. Additionally, this feature would have allowed for more flexible and user-friendly operation, particularly in a manufacturing environment where coil specifications frequently change.

However, due to time constraints in the final stages of development, this feature was ultimately removed from the project scope. While the PLC logic and robotic motion were completed and tested, integrating the spreadsheet interface would have required additional programming, testing, and validation time that was no longer available. Despite being left out, this concept remains a beneficial addition for future system iterations and would significantly improve ease of use and adaptability for real-world applications.

Conclusion

This capstone project demonstrates the successful design and implementation of a fully automated inductor winding system. This project aimed to highlight and solve the issue of the lack of an automated process from the customer's standpoint. It has been built to reflect real-world industrial automation standards and address manual coil production inefficiencies. The project's primary goal was to establish a reliable and repeatable winding workflow that maintains precise coil dimensions and electrical specifications. This system accommodates a continuous supply of copper wire while only needing human intervention to manage spool changes and finished coil retrieval after the process is complete. This setup is designed to efficiently produce wound inductors with consistent quality, reducing manual intervention and operator fatigue. This system combines robotic motion, PLCs, and real-time monitoring safety systems to create a functional, intelligent workstation for precise coil winding. Each component performed its role in coordination, resulting in a system capable of winding inductors of varying specifications with minimal human involvement. One of the problems we encountered was brought to light in the inconsistent and troublesome nature of the winding process. The response to the client's needs was to develop a quick, repeatable, labor-free automated process. With real-time monitoring and a vision system, faults in the process can be quickly detected and rectified if human intervention is necessary. This project achieved its intended technical goals and served as a platform for expanding the team's understanding of industrial automation. With the development of this system, more effort can be placed elsewhere in a bigger manufacturing process, which opens avenues for similar systems on a bigger scale. The hope is that this inductor winder simplifies and

streamlines production for the consumer and reduces operation costs in the long run by saving time and labour in a fast-paced industry where it is crucial. This project lays the foundation for scalable, innovative coil manufacturing solutions in fast-paced, cost-sensitive sectors.

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<https://catalog.orientalmotor.com/item/azd-closed-loop-absolute-type-drivers-ac/azd-series-ethernet-ip-network-driver/azd-aep>

Appendices



Figure 5. Cell Layout

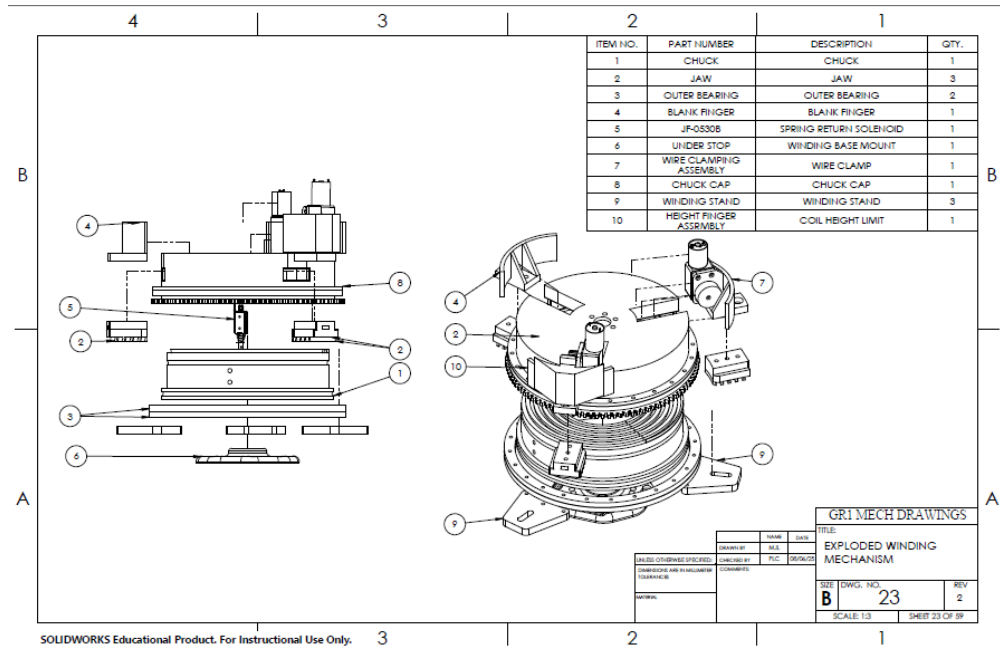


Figure 6. Winding Mechanism

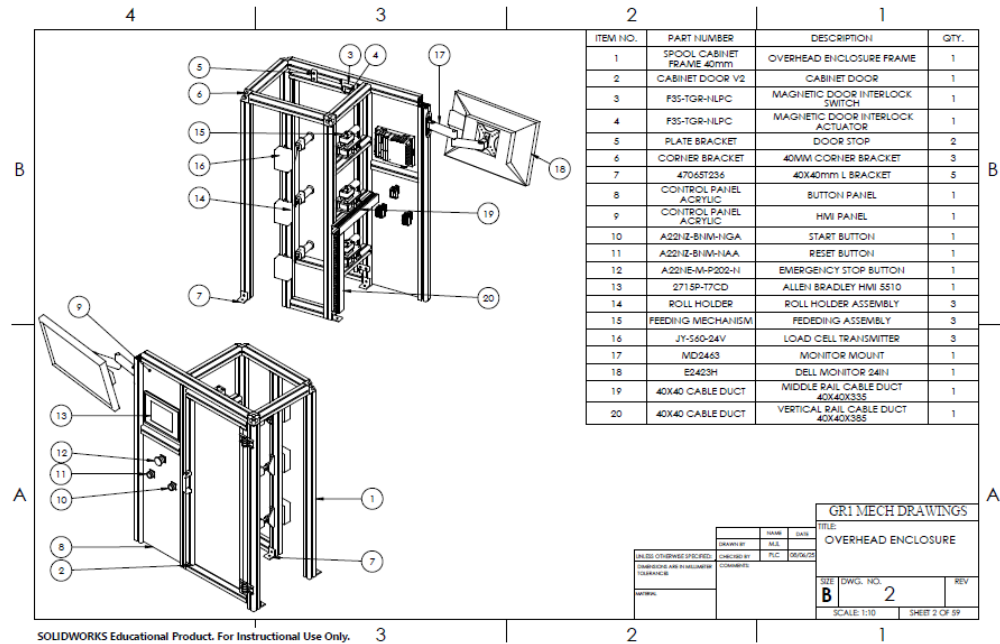


Figure 7. Spool Cabinet

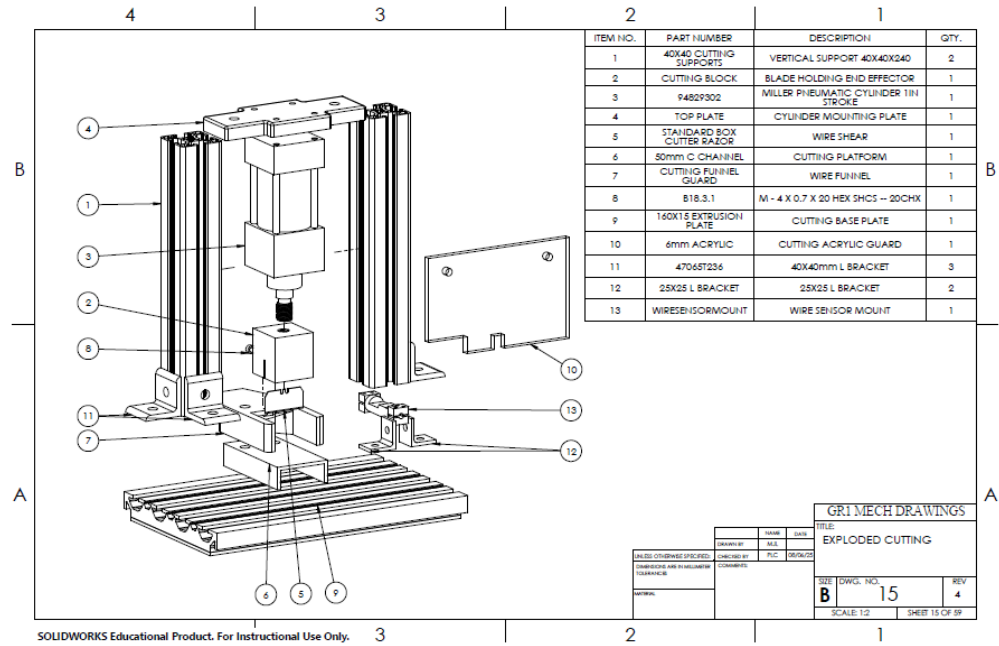


Figure 8. Cutting Mechanism

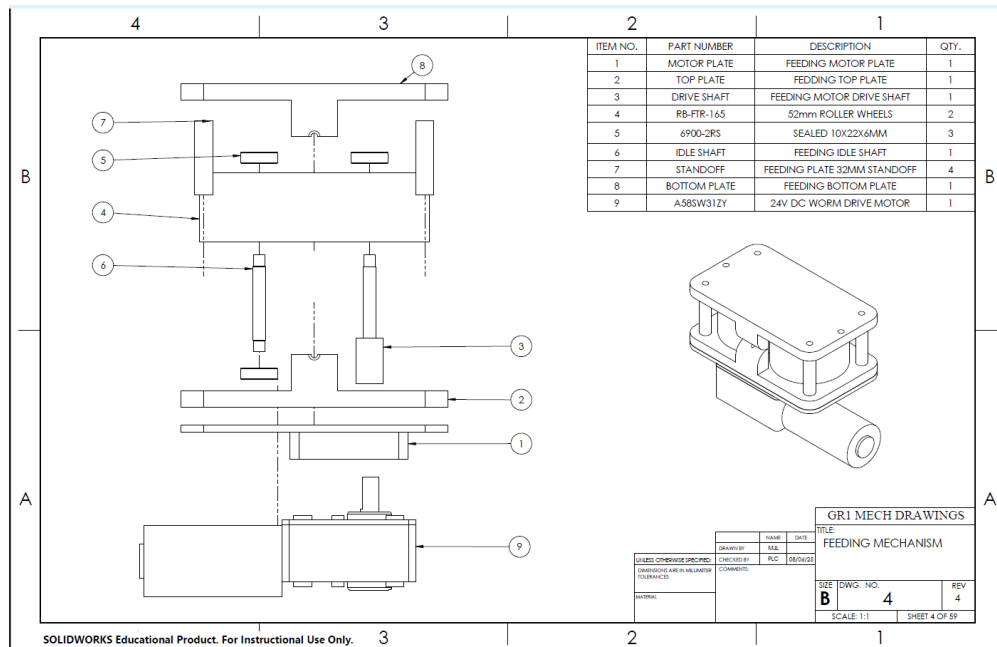


Figure 9. Feeding Mechanism

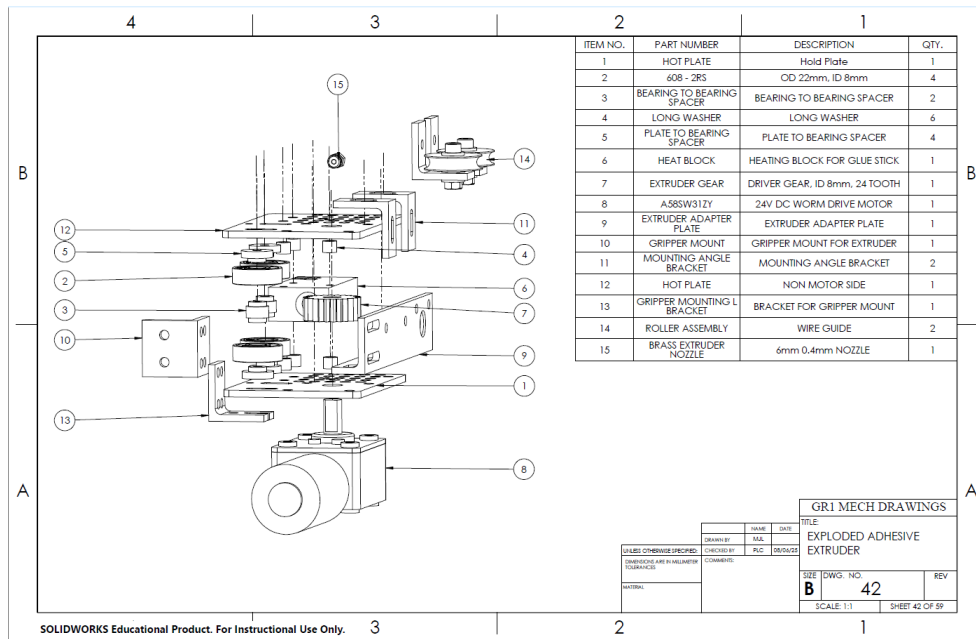


Figure 10. Hot Glue Extruder

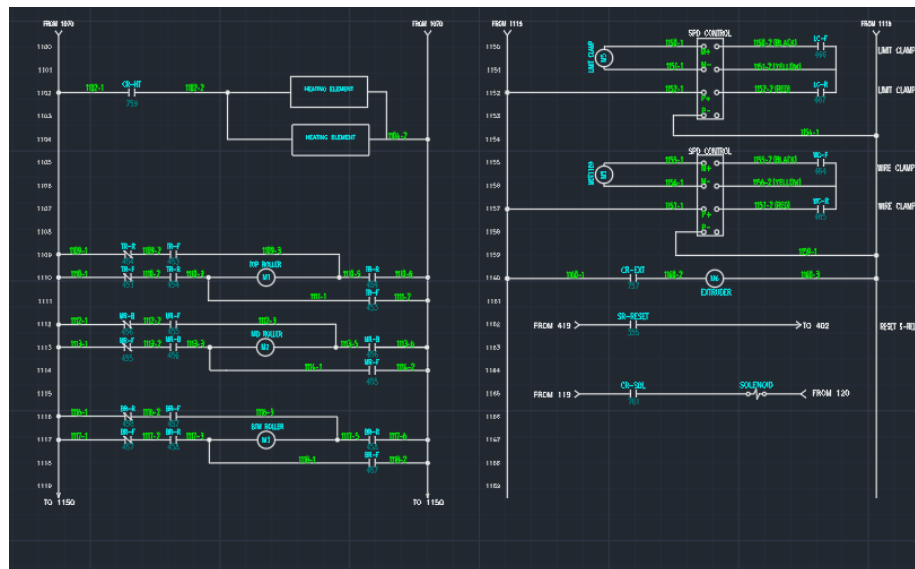


Figure 11. DC Motor Wiring

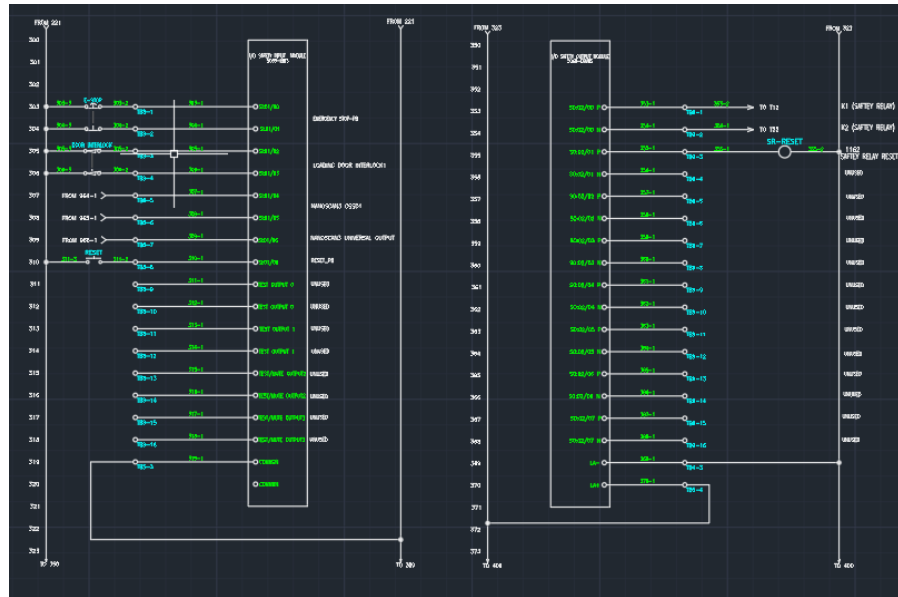


Figure 12. Safety I/O